

**TURNING WASTE INTO WATTS
FROM FIELD TO FUEL**

ENERSOL BIOPOWER PVT. LTD.

Founded in 2013 by **Mr. Rai Singh Dahiya**, a pioneering innovator in biomass gasifier technology, our company is renowned for developing small-scale biomass /coal / RDF gasifiers, smokeless biomass stoves, and other sustainable energy solutions. Committed to innovation and sustainability, we provide efficient and eco-friendly products that promote renewable energy adoption.

Revised & Updated Edition 2025-26

ESB-R12 COAL/BIOMASS GASIFIER

**12kW/15kVA
COAL/BIOMASS GASIFIER SET**



ESB-R5 12 BIOMASS GASIFIER DESIGN

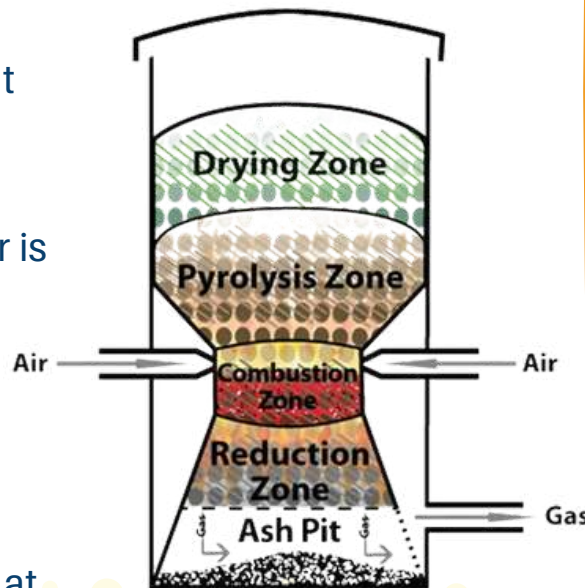


The ESB-R12 12kW coal/ biomass gasifier is ideal for **research institutes and catering to small-scale electricity requirements**. Featuring patented technology, the ESB-R12 biomass/ coal/ coal-biomass gasifier is designed to be compact, user-friendly, and low maintenance, making it a convenient solution for various applications.

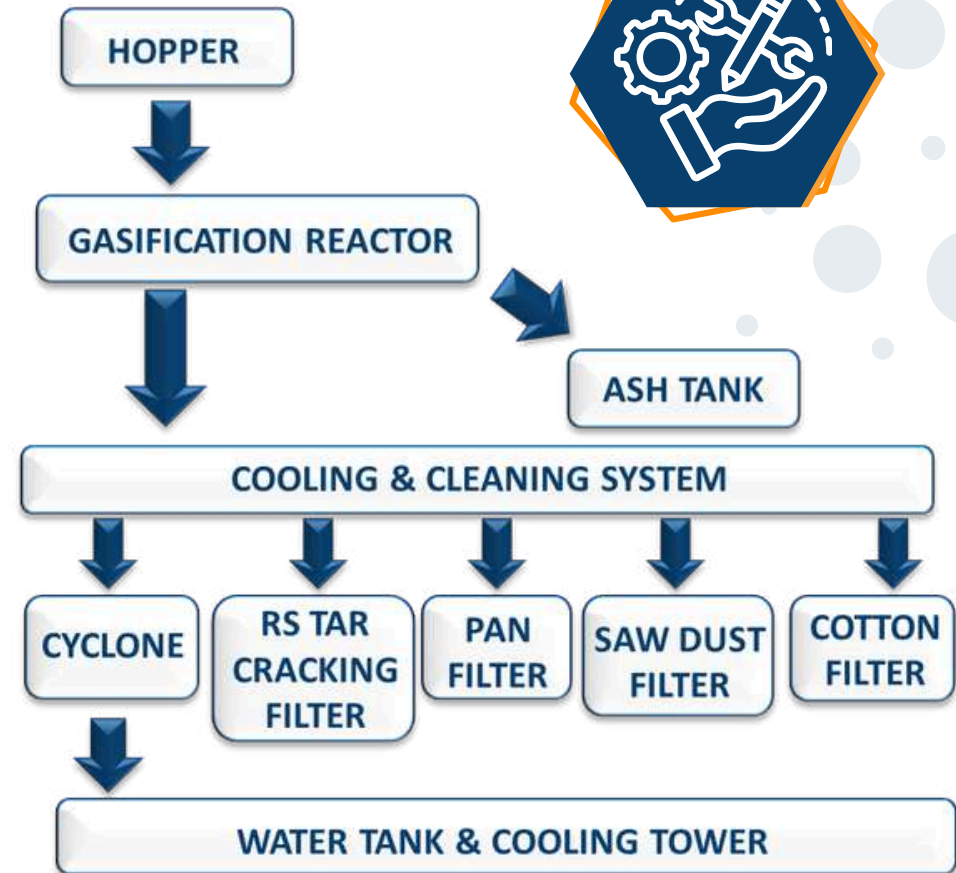
TECHNOLOGY- DOWNDRAFT

KEY FEATURES

- Simple and efficient design.
- Produces low tar content, as most tar is burned in the combustion zone.
- Ideal for use in gas engines.
- Generates fewer particles in the gas at lower loads.



BASIC STEPS INVOLVED



KEY FEATURES OF ESB-R12



The 12kW ESB-R Biomass Gasifier is specially designed to empower small villages by providing sustainable and independent energy solutions. Here are its key features:

- ✓ **Sustainable & Renewable Energy Source**
 - Utilizes biomass like wood chips and agricultural waste, reducing reliance on fossil fuels.
- ✓ **High Efficiency & Reliable Power Output**
 - Generates 50kW of electricity, enough to power small communities and essential services.
- ✓ **Advanced Gasification Technology**
 - Uses a low-tar gasification process, ensuring efficient and eco-friendly energy production.
- ✓ **Off-Grid & Decentralized Energy**
 - Supports standalone microgrids, enabling villages to become self-sufficient in power.
- ✓ **Low Maintenance & Easy Operation**
 - Designed for simple operation, requiring minimal technical expertise and upkeep.
- ✓ **Economic & Environmental Benefits**
 - Reduces diesel dependency, lowers emissions, and creates local employment opportunities.
- ✓ **Scalable & Adaptable Design**
 - Can be integrated with solar or wind energy, making it a flexible solution for rural electrification.



**Most reliable
& rugged
system**



**Compact
system**



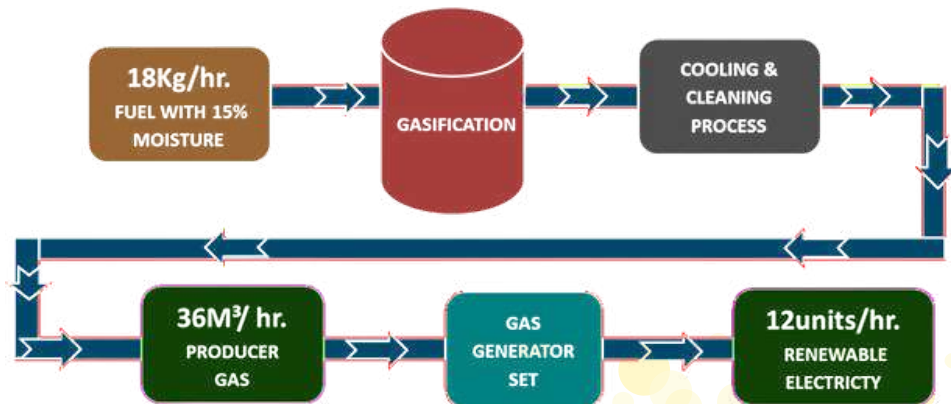
**Accepts all
kind of
biomass / coal
/ RDF**



BASIC PARAMETERS- ESB-R12



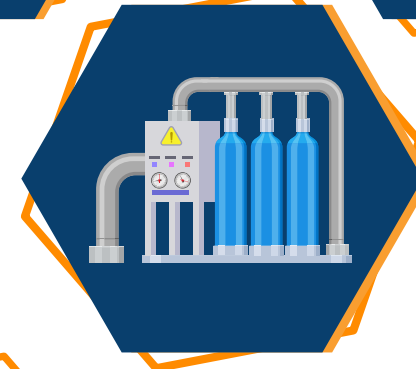
- ✓ ESB-R Biomass Gasifier can operate at any type of agricultural or forest waste residues, North East coal / RDF with limited moisture in feedstock.
- ✓ The gasifier works with producer gas which is basically manufactured 'ON-LINE'.
- ✓ Clean all the filters and do the required maintenance on time for better performance of gasifier.
- ✓ Since Gasifier being a new technology equipment, we advise clients to get properly acquainted with all operational aspects & optional items of Gasifier like biomass / coal characteristic (recommended moisture level, its bulk density, size etc.) as well as suitability of optional items like wood dryer, wood cutter etc. and any other associated accessories.



PATENTED RS FILTER

SELF-GENERATED WATER FOR CLEANING

EFFICIENT TAR CRACKING



CO2 EMISSION REDUCTION

EXTENDED FILTER LIFESPAN

TECHNICAL SPECIFICATIONS



Model Name
ESB-R BIO-12

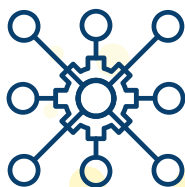
Model NO.
ESB-R12

**Auxiliary
Power**
2.5HP

Weight
2Ton

**Dimension
(Gasifier)**
**L10 x W7 x H8
(feet)**

**Dimension
(Genset)**
**L6x W3.5x H6
(feet)**



BASIC PARAMETERS

Brand	- Enersol Biopower®
✓ Gasifier Capacity	- 15-18kg/hr.
✓ Type of Gasifier	- Downdraft
✓ Gasification efficiency	- >80%
✓ Gasification temperature	- 1000 - 1100°C
✓ Initial Ignition System	- Automatic through Electric Heater
✓ Manpower Required	- 1no
✓ Auxiliary Power Details	- Cooling fan, Vibrator motor, Water pump, Distillation Pump, Conveyor motor, RS filter pump

Parameters	- Details
✓ Acceptable Fuel	- Biomass, North-East Indian Coal, Coal-biomass blend
✓ Coal Type	- Natural non-coking coal (North-East Indian Coal)
✓ Fuel Size	- 1-2mm (Pellets) (fuel dependent)
✓ Coal-Biomass Blend Ratio	- 20:80 to 50:50 (by weight)
✓ Biomass Moisture Content	- <20%
✓ Coal Moisture Content	- <10%
✓ NCV (Biomass)	- 2,100-3,000 Kcal/kg
✓ NCV (Coal)	- 4,000-5,000Kcal/kg
✓ Bulk Density	- =200kg/m ³

TECHNICAL SPECIFICATIONS



Parameters	-	Details
✓ Air Blower System	-	Provided with adjustable air valve
✓ Thermocouples	-	5 Nos. provided for monitoring temperature at various gasifier zones.
✓ Provision for monitoring exhaust gas temperature, pressure and flue gas composition	-	Provided for monitoring exhaust gas temperature, pressure, and flue gas composition (CO, CO ₂ , CH ₄ , H ₂ , SO ₂ , H ₂ S). Monitoring instruments are not included in the scope of supply.
✓ Ash tank	-	Provided

GASIFIER OUTPUT DETAILS

✓ Fuel Consumption (Max)	-	Up to 18 kg/hr. (fuel-dependent)
✓ Producer Gas Flow	-	~36 Nm ³ /hr
✓ Peak Thermal Output	-	38,000 – 45,000 kcal/hr
✓ Continuous Thermal Output (80%)	-	30,000 – 36,000 kcal/hr
✓ Gas Calorific Value	-	= 1,000 kcal/Nm ³
✓ Gasification Temperature	-	1,000 – 1,100 °C
✓ Gas Outlet Temperature	-	300 – 500 °C
✓ Gasification Efficiency (Hot Gas)	-	> 80%
✓ Gasification Efficiency (Cold Gas)	-	> 70%

GAS COMPOSITION

GAS NAME	AVERAGE PERCENTAGE	IN ESB-R12
✓ Carbon dioxide (CO ₂)	10-15%	10.2%
✓ Hydrogen (H ₂)	15-20%	18.9%
✓ Carbon monoxide (CO)	20-25%	21.4%
✓ Methane (CH ₄)	2-5%	3.4%
✓ Nitrogen (N ₂)	40-50%	46%



Carbon monoxide (CO)
21.4%



Hydrogen (H₂) - 18.9%



Nitrogen (N₂) - 46%



Methane (CH₄) - 3.4%



Carbon dioxide (CO₂) - 10.2%

TECHNICAL SPECIFICATIONS



FLUE GAS FLOW CONTROLLER

Parameters	-	Details
✓ Application	-	Producer / syngas
✓ Meter Range	-	3.7 - 254.4 Sm ³ /hr.
✓ Operating Temperature	-	=40°C
✓ Operating Pressure	-	=2 Kg/cm ²
✓ Sensor Type	-	SS dual RTD sensor
✓ Sensor Specification	-	RTD PT20 & PT300/PT1000
✓ Body Pipe Material	-	SS304
✓ Meter Display	-	Graphical LCD Display
✓ Communication	-	RS 485 Modbus RTU
✓ Accuracy	-	+/- 2% of M.V + 0.5% of full scale
✓ Monitoring & data logging	-	Provided



TECHNICAL SPECIFICATIONS

MAJOR COMPONENTS USED IN GASIFIER & ENGINE SYSTEM AND THEIR BRANDS

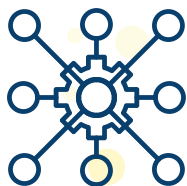
✓ Fan	-	Khetan /Crompton
✓ Electric Motors	-	Havells
✓ Pump	-	Crompton
✓ Control Panel	-	Schneider Electric
✓ Electronic Speed Governor	-	SEDEMAC
✓ Contactor & MCB	-	Schneider Electric
✓ Thermocouples Scanner	-	Multispan / Swastik
✓ Battery (12-24 V)	-	Exide
✓ Cables & Wires	-	Polycab / Havells
✓ Bearings	-	SKF / NBC
✓ Blower	-	Enersol Biopower



COOLING & CLEANING SYSTEM



Parameters	-	Details
✓ Cyclone Separator	-	Removes larger particulate matter and dust from producer gas through centrifugal action.
✓ Scrubber	-	Cools the gas and removes tar, fine particulates, and soluble impurities using water contact.
✓ Pan Filter	-	Eliminates suspended impurities and residual tar droplets from the gas stream.
✓ RS Filter	-	Provides secondary filtration to remove finer particulates and moisture traces
✓ Distillation Tank	-	Separates tar and water from the scrubbing liquid for reuse and proper disposal.
✓ Inbuilt Water Tank & Cooling Tower	-	Supplies and circulates cooling water to maintain desired gas temperature and system efficiency.
✓ Sawdust Filter	-	Absorbs moisture and traps fine dust and tar residues from the gas.
✓ Final Cotton Filter	-	Ensures final stage fine filtration, delivering clean gas suitable for engine/application use.



FILTER CLEANING DETAILS

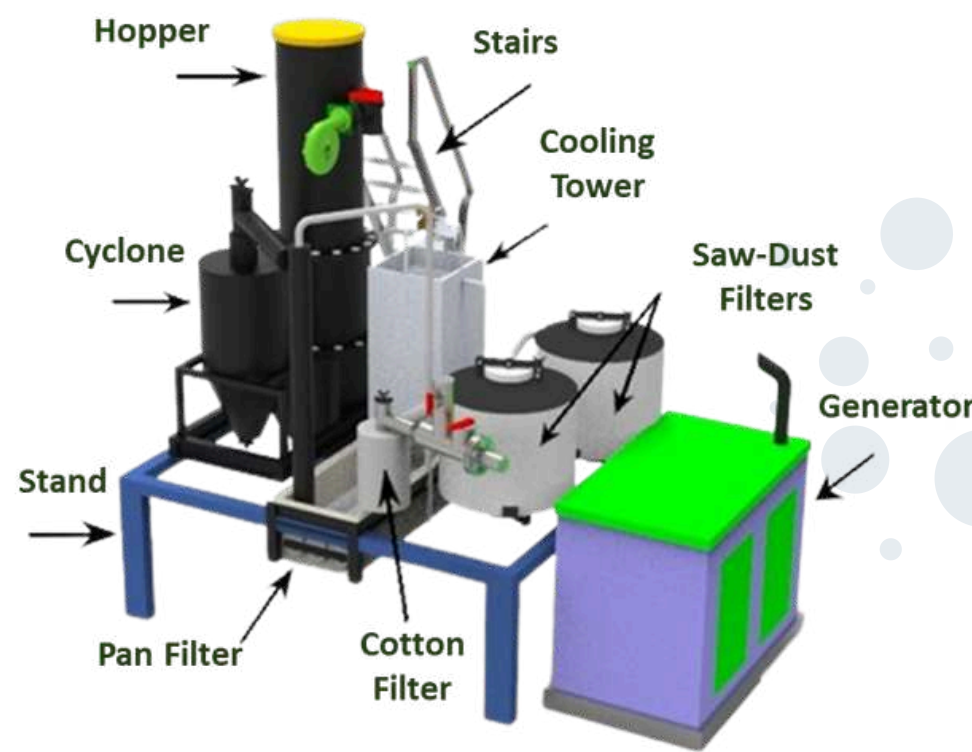
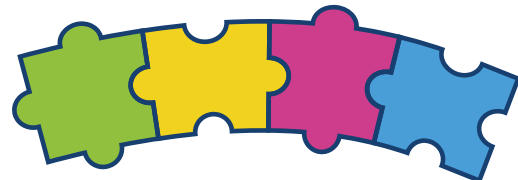
Filter	Cleaning Interval	Remark
✓ Cyclone	Every 40–50 hrs	Ash removal
✓ Scrubber	Every 25–30 hrs	Water quality check
✓ Sawdust / Media Filter	Every 80–100 hrs	Media replacement
✓ Distillation Tank	Every 2–3 hrs	Excess water removal
✓ Pan Filter	Every 80–100 hrs	Cleaning
✓ RS Filter	As required	Inspection
✓ Cotton Filter	Every 80–100 hrs	Fabric replacement
✓ Tar Filter	Every 25-30hrs.	Drain tar every 25–30 hours.
✓ Water Tank Capacity	Approx. 250 L	
✓ Water Level	Maintain a minimum of 75% water level at all times	
✓ Water Evaporation	5–7 L/hr (approx.), depends on climatic conditions	



12KW BIOMASS GASIFIER PARTS



Component	Material
✓ Hopper	- Mild Steel (MS)
✓ Reduction Chamber	- MS+ Ceramic lining
✓ Grate	- High-temperature alloy steel
✓ Ash Storage Tank	- Mild Steel (MS)
✓ Ash Removal Tank	- Mild Steel (MS)
✓ Tar collector tank	- Stainless Steel (SS)
✓ Cyclone	- Mild Steel (MS)
✓ Scrubber	- Stainless Steel (SS)
✓ RS Filter	- SS+ MS
✓ Distillation Tank	- Stainless Steel (SS)
✓ Saw-dust filter	- Stainless Steel (SS)
✓ Pan filter	- Stainless Steel (SS)
✓ Cotton filter	- Stainless Steel (SS)
✓ Water tank	- Stainless Steel (SS)
✓ Coolingtower	- Aluminum with PVC fills



TECHNICAL DETAILS OF GENSET



Parameters	- Details
✓ Engine type	- Producer /syngas
✓ Fuel	- 100% Producer Gas derived from Biomass / Coal / Biomass–Coal Blend
✓ Rated Power Output	- Suitable for continuous operation with 15 kVA alternator
✓ Number of Cylinders	- 2 Nos.
✓ Engine Configuration	- Inline
✓ Stroke	- Four-stroke
✓ Bore x Stroke	- 115mm x 115mm
✓ Compression Ratio	- Optimized for low-CV producer gas
✓ Rated RPM	- 1500RPM
✓ Engine HP	- 16HP
✓ Speed Governor	- Electronic governor Brand - SEDEMAC
✓ Cooling System	- Water-cooled with radiator & cooling fan
✓ Lubrication System	- Forced lubrication with oil pump
✓ Lube Oil Sump Capacity	- 7.2L
✓ Starting System	- Electric Start
✓ Starting Voltage	- 12V DC
✓ Battery	- 12V Exide / equivalent (supplied)
✓ Air Intake System	- Heavy-duty air filter suitable for dusty conditions
✓ Gas Mixing System	- Venturi-based gas-air mixer with fine adjustment
✓ Ignition System	- Electronic Ignition
✓ Exhaust System	- Industrial exhaust with silencer
✓ Engine Mounting	- Anti-vibration mounting

BASIC DETAILS

Model Name
ESB-R
Blo-Gen15

Model No.
ESB-RBG15

Capacity
15kVA
(12kW)

Producer Gas Consumption
36M3/hr.

Ignition
Spark Ignition

Engine rated power
14-16 HP



ALTERNATOR SPECIFICATIONS



Particular	Details
✓ Alternator Make	Enersol/ Topland
✓ Rated Capacity	15 kVA / 12 kW
✓ Phase	Three Phase
✓ Voltage	415 V $\pm$ 1%
✓ Frequency	50 Hz
✓ Power Factor	0.8 lagging
✓ Rated Current	~19 A per phase
✓ Speed	1500 RPM
✓ Insulation Class	Class H
✓ Protection Class	IP 23
✓ Cooling	Air-cooled
✓ Voltage Regulation	$\pm$ 1%
✓ Excitation System	Self-excited
✓ Winding	Copper
✓ Coupling	Direct coupled with engine
✓ Bearings	Heavy-duty, grease lubricated



CONTROL PANEL SPECIFICATIONS

Particular	Details
✓ Control Panel Type	Digital Control Panel
✓ Oil Pressure Gauge	Provided
✓ Water Temperature Gauge	Provided
✓ Hour Meter	Provided
✓ Energy Meter (kWh)	Provided
✓ Voltmeter	Provided
✓ Ammeter	Provided
✓ Frequency Meter	Provided
✓ Phase Indicator Lamps	Provided
✓ Overload Protection	Provided
✓ Short Circuit Protection	Provided
✓ Phase Loss / Phase Reversal Protection	Provided
✓ High / Low Voltage Cut-off	Provided
✓ High / Low Frequency Cut-off	Provided
✓ Battery Charging Alternator	Provided
✓ Emergency Stop Switch	Provided



ACOUSTIC ENCLOSURE & MOUNTING



Particular	Details
✓ Acoustic Enclosure	Soundproof canopy provided
✓ Noise Level	= 75 dB(A) @ 1 m (approx.)
✓ Canopy Material	Powder-coated CRCA sheet
✓ Vibration Isolation	Anti-vibration pads
✓ Base Frame	Heavy-duty MS fabricated skid

MAINTENANCE & SERVICE INTERVALS

Activity	Interval
✓ First lube oil change	50–80 hrs
✓ Second oil change	150–200 hrs
✓ Subsequent oil changes	Every 250–300 hrs
✓ Air filter cleaning	As required
✓ Spark plug inspection	Every 250 hrs

CONTROL PANEL SPECIFICATIONS



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- <https://www.youtube.com/watch?v=OD9ueNYjf1Q>
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- https://www.youtube.com/watch?v=_Ee0_gxwpGc&t=31s
- <https://www.youtube.com/watch?v=fjULT8k4Ayg>
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- <https://www.youtube.com/watch?v=-XmU5L6lytM>
- <https://www.youtube.com/watch?v=2jYnwcBU1lk>
- <https://www.youtube.com/watch?v=LKDlbvXdHyM>
- <https://www.youtube.com/watch?v=7MVHph7NfkY>

OUR PRESTIGIOUS PARTNERS



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AWARDS & EXHIBITIONS

